Oracle-First Native-K/V Materialization for Progressive Retrieval Attention: Separating Memory Discovery from Physical Disclosure

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Abstract

Sparse memory systems often treat finding a relevant region and loading its complete key/value (K/V) payload as one operation. This coupling obscures a separate question: once the relevant conceptual memories are known, how much layer-native detail does a frozen transformer need? We isolate that question in Progressive Retrieval Attention (PRA) by fixing dataset-annotated evidence and varying only physical disclosure. Materialization uses source-relative logical intervals, unions overlap, crosses storage-chunk boundaries, preserves explicit-resource boundaries and grouped-query-attention shapes, and transfers only gathered K/V from CPU to GPU.

Experiments use frozen Qwen3-0.6B on MuSiQue and 2WikiMultiHopQA. Four validation examples per dataset select a radius before an identity-disjoint eight-example held-out evaluation. Exact evidence disclosure preserves the same measured generated-answer F1 and normalized accuracy as whole 32-token parent disclosure while reducing unique K/V by 19.8% on MuSiQue and 53.0% on 2Wiki. The reduction is not free in likelihood: gold-token log probability falls by 0.246 and 0.210 nats/token, respectively, although evidence attention increases. A 128-token cap is 97.5% evidence-dense on MuSiQue but covers only 53.3% of annotated evidence and degrades likelihood; on 2Wiki it covers 96.6% and nearly matches whole-parent likelihood. In a separate frozen Paper-2.5 selector factorial, local disclosure halves the graph-sparse payload on MuSiQue and reduces it 38.2% on 2Wiki; a 128-token cap raises 2Wiki output F1 from .50 to .75 but fails on distributed MuSiQue evidence. Native gist means do not substitute for token detail, and adding them to local detail gives no consistent benefit.

The result is a bounded but useful claim. Discovery breadth and active native K/V need not be identical, and evidence-centered disclosure can reduce softmax competition and physical state. Whether that reduction preserves model confidence depends on evidence distribution and memory-use adaptation. The study is a descriptive pilot, not a serving-speed or significance claim.

1 Introduction

A retrieval system answers at least three questions. Which memory is relevant? Which physical detail should become active? Can the model use that detail to produce the answer? These questions are easy to collapse because ordinary retrieval-augmented generation serializes selected text into one prompt. Once a document is selected, all of its re-tokenized prompt state participates in attention. Native-K/V memory exposes the separation more directly. A compact representation can identify a conceptual region while the attention kernel receives a smaller set of already computed K/V states.

PRA Paper 1 separated logical memory from active attention state [5]. Paper 1.5 showed why semantic routing state and native positional attention state should remain distinct [4]. Paper 2 integrated that boundary into pretrained Hugging Face decoder families [6]. Paper 2.5 then treated discovery as bounded traversal over conceptual memory regions. The present paper asks the next question:

> Once relevant conceptual memory is known, how little native K/V detail must be materialized for a frozen model to preserve useful output behavior?

The primary experiment begins with an oracle. Dataset annotations specify the evidence intervals, so a router cannot take credit or blame for the result. We vary exact evidence, local context, whole routing parents, fixed budgets, native gist means, and gist-plus-local hybrids. Only after freezing the oracle frontier do we cross Paper-2.5's one-shot and graph selectors with the same materialization policies. The experimental identity is therefore

$$Q_{\text{output}} = f(M \mid S), \tag{1}$$

where selection S is fixed within a comparison and materialization M alone changes.

This distinction matters economically and scientifically. Broad conceptual search can improve recall, but whole-parent disclosure makes breadth expensive in K/V and introduces irrelevant softmax competitors. Conversely, a tiny physical budget can be evidence-dense while omitting required evidence regions. We therefore report both evidence density and evidence coverage, rather than treating either as sufficient.

The contributions are:

1. a model-independent logical interval abstraction for layer-native K/V;
2. exact cross-chunk gathering with deduplication, resource isolation, GQA preservation, and CPU-to-GPU transfer accounting;
3. an oracle-first M0–M7 materialization ladder and a predeclared radius/budget protocol;
4. paired output, attention, memory, and latency measurements on MuSiQue and 2Wiki;
5. a frozen selection-by-materialization factorial using Paper-2.5 conceptual selections; and
6. explicit negative results for native gist-only disclosure, hard-budget evidence truncation, frozen MuSiQue memory use, and the absence of a serving-speed result.

2 Three Independent Decisions

2.1 Selection, materialization, and use

Let $\mathcal{C}$ denote conceptual memory regions, $S(q, \mathcal{C})$ a selection procedure, $M(S, B)$ a materializer under physical budget B, and U the frozen transformer's native attention and generation path. PRA computes

$$C_q = S(q, \mathcal{C}), \tag{2}$$

$$(K_M, V_M) = M(C_q, B), \tag{3}$$

$$y = U(q, K_M, V_M). \tag{4}$$

Paper 2.5 studies the first line. This paper freezes the first line and studies the second. A memory-use adapter would change the third line and is outside the primary experiment.

This factorization prevents two misleading conclusions. First, high evidence recall does not prove that whole selected parents help output; their irrelevant K/V can dilute attention. Second, poor output after a small materialization does not prove that selection failed; the selected concept may be correct while its disclosed detail lacks context.

2.2 Three granularities

We record three independent granularities:

$$G_{\text{encode}} \neq G_{\text{search}} \neq G_{\text{materialize}}. \tag{5}$$

Canonical contextual K/V is encoded in model-safe blocks of 256 tokens. Search addresses 32-token parents. Materialization operates on arbitrary source-relative token intervals. Storage boundaries therefore do not define semantic windows.

2.3 Logical materialization domains

An interval is a half-open tuple

$$I = (d, s, e), \qquad 0 \leq s < e, \tag{6}$$

where d is a logical domain. Every explicit URI is a separate domain. A window near the end of one resource cannot continue into another resource merely because their tensors are adjacent in a cache. The displaced long-prompt resource `#__head` and the live prompt tail are one exception: they share the continuous prompt-history domain established by canonical PRA.

For annotated evidence $[s_i, e_i)$, a symmetric local policy requests

$$I_i(r) = [\max(0, s_i - r), \min(L_d, e_i + r)), \tag{7}$$

with clipping inside domain d. The physical request is the domain-wise union

$$\mathcal{I} = \bigcup_i I_i, \tag{8}$$

so each logical token is materialized at most once.

3 Materialization Mechanism

3.1 Stored shards and canonical provenance

Each reference layer stores shards with logical span, source URI, canonical post-position K/V, and encoding provenance. Primary experiments never re-encode isolated evidence. If a semantic window crosses two routing chunks, the gatherer slices both stored shards and concatenates them in logical order. For grouped-query attention, K/V remain

$$[1, H_{kv}, T, D_h], \tag{9}$$

without expansion to query-head count.

Figure 1 shows the separation. Conceptual selection stops before the physical materializer. The direct prompt K/V and selected memory K/V meet only at native attention.

logical source / long prompt → model-safe contextual encoding blocks → canonical native K/V shards

↓

smaller routing parents → gists / packed conceptual index → fixed oracle or learned selection

↓

logical interval planner → domain union and deduplication → cross-shard gather under K/V budget

↓

selected native K/V + direct prompt K/V → pretrained native attention

Figure 1: Selection and physical disclosure are separate operations. Search parents are not required to be encoding blocks or materialization windows.

3.2 Cross-shard gather

For each merged interval, the resolver advances a logical cursor. Among overlapping storage shards covering the cursor, it selects the shard extending furthest right, slices only the uncovered range, and repeats until the interval ends. Missing coverage raises an error instead of borrowing tokens from another URI. CPU-resident slices are collected before transfer; only the selected slices move to the GPU.

Listing 1: Simplified logical K/V gathering.

```
merged = union_intervals(requested_intervals)
for interval in merged:
    cursor = interval.start
    while cursor < interval.end:
        shard = furthest_shard_covering(interval.domain, cursor)
        stop = min(interval.end, shard.end)
        k_parts.append(shard.k[:, :, cursor-shard.start:stop-shard.start, :])
        v_parts.append(shard.v[:, :, cursor-shard.start:stop-shard.start, :])
        cursor = stop
K = torch.cat([part.to(device) for part in k_parts], dim=2)
V = torch.cat([part.to(device) for part in v_parts], dim=2)
```

The implementation records requested tokens before union, deduplicated tokens, materialized bytes, evidence and non-evidence tokens, shards touched, cross-shard interval count, interval-resolution time, gather time, and host-to-device time.

3.3 Fixed budgets across evidence regions

For n regions with capacities c_i, a fixed budget assigns integer quotas b_i satisfying

$$1 \leq b_i \leq c_i, \qquad \sum_i b_i \leq B. \tag{10}$$

Equal, evidence-length-proportional, minimum-core-plus-remainder, and score-proportional allocation are implemented. Gate 1 uses equal allocation. Every region receives at least one token, but a budget smaller than the complete evidence cores may explicitly truncate those cores. Evidence coverage exposes that truncation; it is never inferred from density.

4 Oracle Policy Ladder

Table 1 defines the non-learned ladder. M4 is not a separate tensor path: deduplication is mandatory for every interval policy. M6 first forms radius-64 candidates and then allocates the fixed physical budget.

Table 1: Oracle-first disclosure policies.

Policy	Name	Physical disclosure
M0	native gist only	One native-head mean K/V state per selected parent
M1	whole parent	Every K/V token in each selected 32-token routing parent
M2	evidence only	Exact union of annotated evidence token intervals
M3	local window	Symmetric radii $r \in \{0, 4, 8, 16, 32, 64\}$
M4	deduplicated windows	Domain-wise union before physical gather; active in M2–M7
M5	asymmetric window	Eight tokens left and 32 tokens right of evidence
M6	fixed budget	Equal allocation at 64, 128, and 256 native K/V tokens
M7	gist plus local	Native K/V mean for every selected parent plus local token K/V

The native gist in M0/M7 is deliberately not the routing vector. On a GQA model, routing width can equal hidden size while native K/V width is $H_{kv}D_h$. We instead average each parent's actual native K and V independently over tokens, preserving the physical head layout.

5 Experimental Protocol

5.1 Model, data, and frozen controls

Table 2 summarizes the experiment. MuSiQue and 2Wiki retain their annotation graphs and exact serialized source spans from Paper 2.5. Validation and held-out identities are disjoint. No optimization occurs in this paper.

Table 2: Frozen oracle-frontier protocol.

Property	Value
Backbone	Qwen3-0.6B, frozen, eager native attention, FP16 CUDA
Datasets	MuSiQue and 2WikiMultiHopQA annotated development examples
Selection	Dataset evidence oracle for Gate 1; frozen Paper-2.5 selectors for Gate 2
Validation / held-out	4 / 8 identity-disjoint examples per dataset
Selector factorial	4 held-out examples per dataset, 4 selectors × 4 policies
G_{encode}/G_{search}	256 / 32 source tokens
Materialization layers	Inherited Paper-2.5 bands: all 28 for MuSiQue; layers 12,20,24,27 for 2Wiki
Direct prompt / native limit	128 / 4096 tokens
Decode	Greedy, at most 12 new tokens; brief-answer extraction
Validation rule	Smallest radius within 0.05 nats/token of whole-parent likelihood; otherwise best likelihood
Hardware	NVIDIA GeForce GTX 950M, 4 GB; reference K/V resident on CPU

The validation rule selects radius zero on both datasets. Held-out evaluation therefore compares M2 evidence only without revisiting the radius choice. We report exact match, token F1, containment, normalized answer accuracy, gold answer log probability, attention mass, K/V tokens and bytes, evidence density and coverage, peak CUDA memory, TTFT, TPOT, and gather/transfer timing.

The optional dense full-context control exceeded the 4 GB GPU during eager attention and is omitted. This does not remove the whole-parent PRA ceiling, but it prevents a direct full-context comparison. Timing is synchronized single-example research timing; it is not a loaded-serving benchmark.

6 Oracle Frontier Results

6.1 Validation selects exact evidence

Figure 2 shows the validation frontier. Increasing radius does not monotonically improve likelihood. Exact evidence is the strongest measured point on both validation slices. On MuSiQue it uses 350.5 rather than 415.5 unique K/V tokens and improves likelihood from −9.073 to −8.281. On 2Wiki it uses 82.8 rather than 144.0 tokens and improves likelihood from −4.716 to −4.354. These four-example values select the policy; they are not the final estimate.

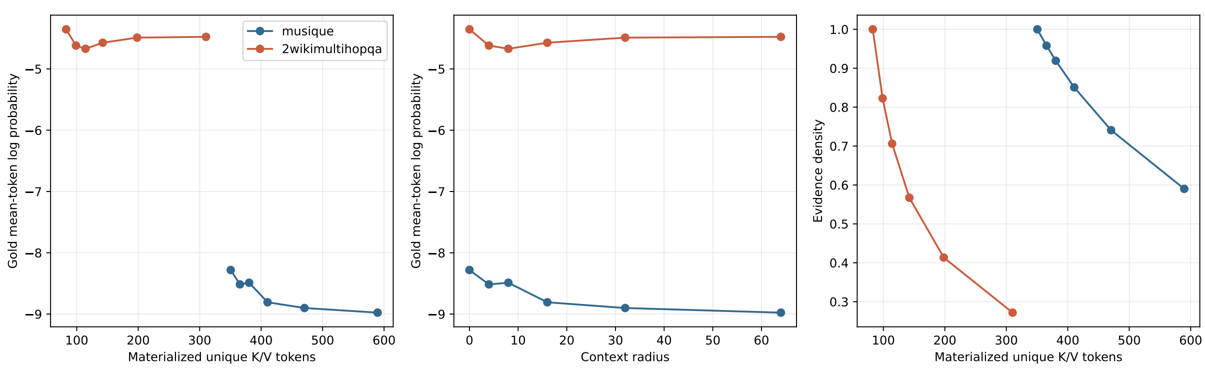

Figure 2: Validation quality/K/V frontier, quality versus radius, and evidence density. Radius zero is selected before held-out evaluation.

6.2 Held-out quality and K/V

Table 3 gives the held-out result. Exact evidence preserves the same generated answer metrics as whole parents on both datasets. This is useful but should not be overstated: MuSiQue F1 is zero for both policies, while 2Wiki F1 and normalized accuracy are .50. Likelihood detects a cost hidden by those coarse output metrics.

Table 3: Held-out oracle disclosure, eight examples per dataset. LP is mean gold-token log probability. Coverage is fraction of annotated evidence physically present.

Dataset	Policy	K/V	Reduction	Density	Coverage	LP	F1
MuSiQue	whole parent	313.5	0.0%	.802	1.000	−8.774	.000
MuSiQue	evidence only	253.1	19.8%	1.000	1.000	−9.020	.000
MuSiQue	budget 128	128.0	57.2%	.975	.533	−9.542	.000
2Wiki	whole parent	136.0	0.0%	.470	1.000	−5.528	.500
2Wiki	evidence only	62.5	53.0%	1.000	1.000	−5.738	.500
2Wiki	budget 128	124.9	−6.7%	.476	.966	−5.551	.500

Evidence-only likelihood falls 0.246 nats/token on MuSiQue and 0.210 on 2Wiki relative to whole parents. It wins or ties on three of eight and two of eight paired examples, respectively. The result supports K/V compression at fixed deterministic output, not likelihood equivalence.

6.3 Attention dilution

The smaller disclosure concentrates attention. Evidence attention rises from .522 to .662 on MuSiQue and from .161 to .194 on 2Wiki. Thus less irrelevant K/V produces more evidence-directed attention, even when likelihood does not improve. This separates a mechanistic effect from an output claim: removing softmax competitors is measurable, but the frozen model may still need surrounding context or adaptation to interpret the remaining states.

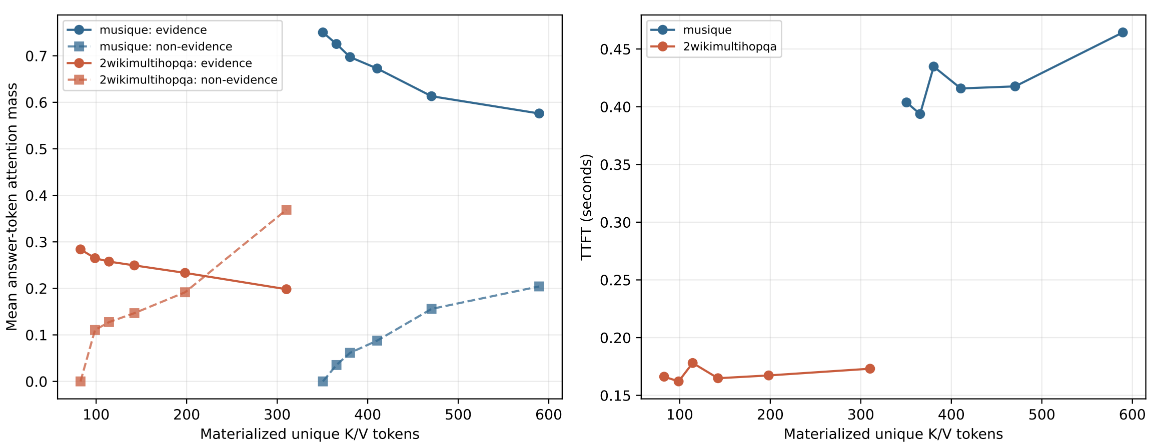

Figure 3: Validation evidence/non-evidence attention and synchronized TTFT against materialized K/V. The small pilot does not support a serving-speed claim.

6.4 Density is not coverage

The fixed-budget result is the clearest warning. MuSiQue budget-128 disclosure is 97.5% evidence-dense, apparently an efficient payload, but it includes only 53.3% of the annotated evidence core. Its likelihood loses another 0.522 nats/token relative to exact evidence. On 2Wiki, the same policy covers 96.6% of evidence and is only 0.023 nats/token below whole parents. The parameter is identical; evidence distribution differs.

This result rejects a simple global rule such as “128 K/V tokens are enough.” A fixed budget needs evidence-region-aware feasibility or an adaptive stopping criterion. Minimum one-token allocation prevents silent total starvation, but it cannot preserve a long evidence paragraph.

6.5 Gists are routing state, not sufficient detail

M0 reduces K/V by about 97%, but MuSiQue likelihood is 7.374 nats/token below whole parents and 2Wiki is 0.582 below. M7 adds one native gist per selected parent to exact evidence. It changes MuSiQue likelihood by less than .001 relative to evidence only and loses .007 on 2Wiki, while using more K/V. Mean native K/V therefore does not replace token detail in this configuration and does not justify a hybrid default.

6.6 Frozen memory-use boundary

No-memory likelihood is -4.491 on held-out MuSiQue, substantially above every memory policy. On 2Wiki, whole-parent memory improves over no memory by 0.564 nats/token and evidence-only improves by 0.354. Paper 3 does not solve this dataset-level consumption mismatch because the backbone and memory-use path are frozen. The MuSiQue result is evidence against interpreting an oracle K/V frontier as proof that a pretrained model can use all annotated evidence. The materializer can be correct while the consumer is not calibrated.

7 Frozen Selection by Materialization

After Gate 1, we reuse one-shot, graph-sparse, graph-balanced, and graph-high Paper-2.5 selected spans. Oracle annotations are unavailable to both selection and interval construction. For each selector we compare the selected 32-token parents, the selector's own atomic logical spans, a 128-token equal allocation, and native gist plus atomic detail.

Table 4: Representative selector/materializer pairs, four held-out examples per dataset. Δ LP and ΔF1 are relative to the same selector with whole-parent disclosure.

Dataset	Selector / disclosure	Parents	K/V	Reduction	ΔLP	ΔF1
MuSiQue	graph sparse / local	13.75	216.2	49.4%	-0.153	.000
MuSiQue	graph sparse / budget 128	13.75	128.0	69.8%	-6.690	.000
MuSiQue	graph balanced / budget 128	11.75	128.0	91.4%	-8.273	.000
2Wiki	graph sparse / local	14.00	221.5	38.2%	+0.069	+.250
2Wiki	graph sparse / budget 128	14.00	128.0	63.3%	+0.223	+.250
2Wiki	graph balanced / budget 128	6.00	128.0	81.8%	+0.280	+.250
2Wiki	graph high / budget 128	9.50	128.0	87.8%	+0.427	+.250

Graph-sparse selection supplies the cleanest direct test because its selected atomic spans are finer than their storage parents. Local materialization nearly halves MuSiQue payload with a modest likelihood loss and reduces 2Wiki payload while improving both likelihood and F1. For one-shot, graph-balanced, and graph-high conditions, many selected spans already cover complete 32-token parents or merge into broad contiguous regions, so local materialization is identical to whole-parent disclosure. Multi-scale discovery does not guarantee a finer physical payload if its final selected intervals are coarse.

The hard budget again exposes a dataset interaction. Every reported 2Wiki selector reaches F1 .75 under 128 tokens, compared with .50 for its whole-parent counterpart. MuSiQue remains at the zero-F1 floor and suffers large likelihood losses. Smart materialization can bound broad conceptual discovery on 2Wiki, but a common global budget does not yet handle MuSiQue's distributed multi-region evidence.

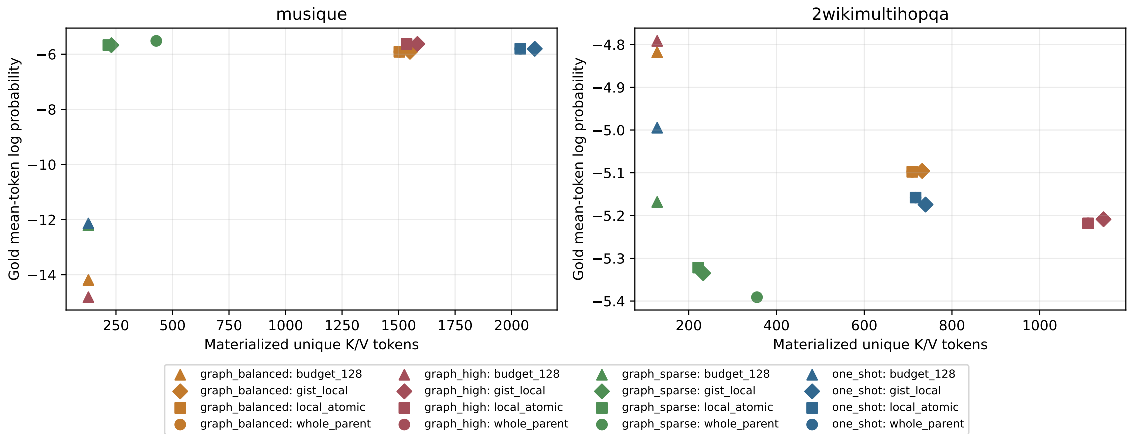

Figure 4: Frozen selection-by-materialization factorial. The same conceptual selection can occupy different physical K/V points; broad selected spans sometimes leave no local compression room.

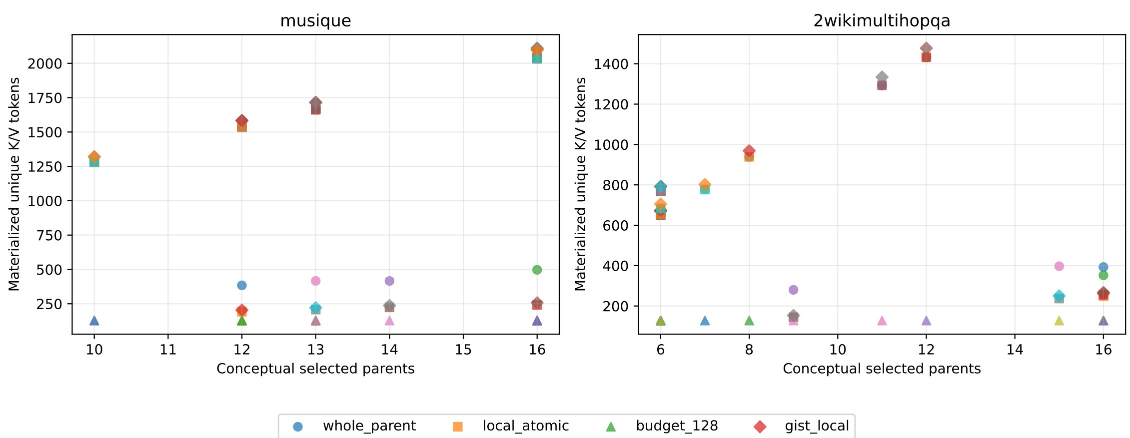

Figure 5: Conceptual parent count versus physical K/V. Fixed budgets decouple the axes, but quality depends on whether the budget preserves all required evidence regions.

8 Systems Accounting

Reference K/V remains on CPU in native KV-head form. The materializer slices CPU tensors before transfer and records final H2D bytes/time. On the held-out oracle evidence condition, summed cross-shard interval counts are 63 per MuSiQue example across 28 active layers and 7 per 2Wiki example across four active layers. Thus semantic evidence commonly crosses storage boundaries; clipping at the selected parent would change the experiment.

Deduplication also has observable effect. Held-out 2Wiki evidence requests average 65.1 tokens before union and materialize 62.5, a 4.0% reduction. Validation radius 64 requests 338.8 and materializes 310.3, an 8.4% reduction. MuSiQue's selected examples contain little interval overlap, so deduplication saves little there. Deduplication is semantically required even when it does not dominate runtime.

TTFT does not monotonically follow selected K/V in this small synchronized experiment. Evidence only is 0.080 seconds slower than whole parents on MuSiQue and 0.008 seconds slower on 2Wiki. Python interval resolution, many small slices, eager attention, Windows WDDM, and single-example noise dominate the token reduction. The result is a memory frontier, not a speed frontier. Fused interval indexing and batched gathers belong to a systems follow-up.

9 Implementation and Reproducibility

The implementation is split at the same conceptual boundaries as the method. In the revision used for this paper:

- `src/pra_torch/materialization.py:35` defines logical intervals;
- line 81 defines provenance-bearing K/V shards;
- line 225 performs domain-wise interval union;
- line 312 allocates fixed budgets across regions;
- lines 378–401 resolve exact cross-shard coverage; and
- line 403 gathers ordered K/V and records physical/transfer metrics.

The shared execution core parses interval requests at `src/pra_torch/core.py:211`, materializes them at line 249, constructs native gist controls at line 359, and dispatches all legacy and Paper-3 disclosure modes through the same fixed-selection boundary at line 449. The public and internal HF configurations expose `detail_materialization`; reference construction records encoding granularity in `src/pra_torch/hf/injection.py:446`.

The oracle harness, selector factorial, and analysis are in `experiments/paper3_kv_materialization`. Canonical JSON/CSV rows and figures are under `docs/papers/shared/results/paper3_kv_materialization`. Each experiment writes a JSONL checkpoint after every condition. Dataset annotations enter Gate 1 interval planning but are explicitly unavailable to Gate 2 selectors and materializers.

10 Related Work

Sparse-attention architectures reduce the token pairs processed inside a model through fixed or learned patterns [1]. Recurrent and compressed-context approaches retain history through cache policies or summary state; StreamingLLM studies stable attention sinks for streaming caches [8], while Infini-attention introduces compressive memory inside attention [3]. Memorizing Transformers retrieve approximate nearest-neighbor K/V from prior examples [7], and landmark attention separates block-level addressing from token-level access [2]. These systems differ in training and storage contracts, but they motivate separating compact addressing from detailed state.

PRA's materialization question is narrower. The selected payload is layer-native contextual K/V, the host backbone is frozen, and the experiment begins with oracle evidence so retrieval recall cannot confound disclosure. The contribution is not a new end-to-end long-context benchmark leader. It is a controlled physical interface and evidence about how output changes along its K/V frontier.

11 Limitations

The held-out oracle study has eight examples per dataset and the selector factorial has four. Results are descriptive; no confidence interval or significance claim is warranted. Greedy exact match and F1 are coarse, particularly on MuSiQue where every compared condition is at the zero-F1 floor. A validated blinded LLM judge has not yet been run for Paper 3.

Only Qwen3-0.6B is evaluated. Materialization behavior may differ with model scale, family, fine-tuning, quantization, and attention kernel. The inherited all-28 MuSiQue materialization band is expensive and may amplify frozen-model interference. We preserve it to avoid selecting a new layer policy on the materialization outcome.

The evidence oracle assumes dataset annotations identify useful source spans. It does not prove that every token in an annotated paragraph is causally necessary. Conversely, exact evidence can lack discourse or entity context outside the annotation. M2's likelihood loss is consistent with that possibility but does not identify the cause by itself.

The direct full-context control OOMed on the available 4 GB GPU under dense eager attention. Latency uses a single Windows GPU without loaded concurrency, and Python gather overhead remains. The paper therefore makes no production throughput, cost, or asymptotic speed claim.

12 Discussion and Paper-3.5 Handoff

The central hypothesis survives in a qualified form. Conceptual discovery breadth and physical native K/V can be decoupled. Exact evidence often occupies substantially less K/V than whole routing parents, increases evidence attention, and preserves the measured greedy answer metrics. The likelihood penalty shows that annotations alone do not specify all useful context. A practical materializer should therefore choose context adaptively rather than assume either whole parents or zero radius.

Three next steps follow directly. First, budget allocation should preserve complete evidence cores before distributing remainder, or explicitly defer a region rather than truncate it. Second, selected native positions should seed asymmetric windows whose size depends on boundary, punctuation, and confidence. Third, materialization and memory-use adaptation should be trained jointly only after retaining the oracle controls; otherwise an adapter can conceal disclosure failure.

Paper 3.5 receives a concrete interface: logical intervals, cross-shard gather, exact metrics, radius-zero and 128-token target points, CPU/GPU transfer profiles, and the observed gather bottleneck. It should own learned adaptive budgets, fused indexing/gather kernels, batching, concurrency, and cross-family serving measurements.

13 Conclusion

Finding memory and loading memory are different operations. Under oracle selection, exact evidence reduces active native K/V and softmax competition while preserving the measured generated answer scores, but it does not preserve gold-token likelihood exactly. Fixed budgets work when they retain the evidence distribution and fail when density hides low coverage. With frozen Paper-2.5 selectors, sparse conceptual discovery benefits most from finer disclosure; broad selected spans may leave nothing to compress. These results establish materialization as its own research object and provide a reproducible boundary for the next systems and learning stage.

A Test Matrix

Focused tests cover intervals inside one shard, one-boundary and multi-boundary gathers, overlapping storage shards, overlapping requested intervals, exact token order, resource start/end clipping, explicit-resource rejection, `#__head`-to-prompt continuity, GQA shape preservation, CPU-to-GPU transfer bytes, fixed budgets, multi-evidence allocation, radius zero, whole-parent equivalence, logical-core metrics, native gist controls, and budget rejection. Existing PRA, Paper-2, and Paper-2.5 regressions run in the full project suite.

B Canonical Row Schema

Every result row records dataset and identity, selection and materialization policy, radii and K/V budget, conceptual parents, intervals, requested and deduplicated tokens, unique tokens, per-layer token states and bytes, evidence/non-evidence tokens, density and coverage, layer band, output and teacher-forced metrics, attention mass, interval/gather/H2D timing, TTFT, TPOT, total generation, throughput, and peak CUDA allocated/reserved bytes. Oracle use is an explicit boolean.

C Negative Result Ledger

Observation	Consequence
Native gist-only collapses quality	Gists remain routing/index state, not a default detail payload.
Evidence-only loses likelihood	Zero-radius evidence is a compression point, not universal equivalence.
MuSiQue budget 128 covers 53.3%	Density must always be paired with evidence coverage.
Frozen MuSiQue replay loses to no memory	Correct materialization does not solve memory-use calibration.
Broad selected spans do not shrink locally	Finer physical disclosure requires finer selected intervals or a within-parent policy.
Python gather does not improve TTFT	Optimize indexed/fused gather before speed or cost claims.
Direct full context OOMs	No dense full-context ceiling is reported on the 4 GB device.
No blinded judge yet	Behavioral equivalence is limited to deterministic and teacher-forced metrics.

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